Sufficient Statistics for Economic Mobility: When Do Measures Agree?

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UWO

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Has U.S. Intergenerational Mobility Declined?

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  - ... in many outcomes (e.g., incomes, wealth, education, health, ...)
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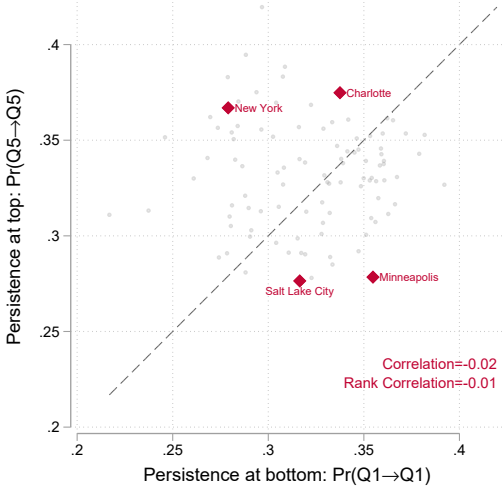
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- But not always
  - ... U.S. over time (mid-century births: sharp ↓ Davis & Mazumder 2024 vs. modest to flat Jácome, Kuziemko & Naidu 2025)
  - ... across countries (Blanden 2013) and across places (Chetty et al., 2014)

Even Positional Mobility Measures Disagree

Persistence across the 100 largest commuting zones (Chetty et al., 2014)

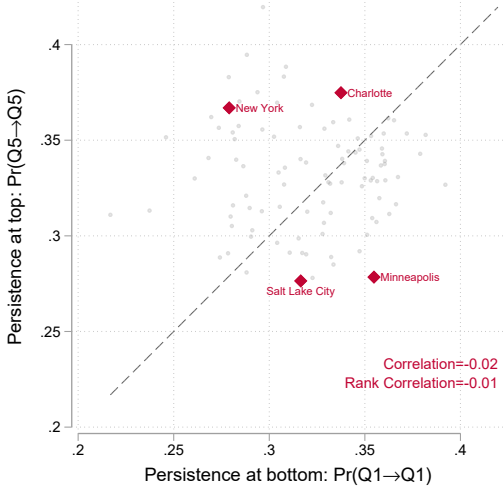
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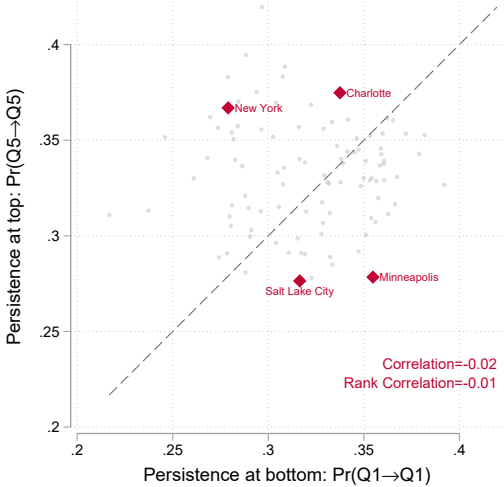
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Two fundamental questions:

1. When *should* we expect measures to agree?
2. What does disagreement *tell us*?



This Paper Provides Answers

1. Agreement of measures depends on single property of parent-child joint distribution
   - Concordance order: parent-child ranks are *everywhere* more aligned
   - If economies ranked by concordance, *all* standard measures agree
     $\longrightarrow$ measures as ordinal substitutes (for transition probabilities, *only* then)

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2. Disagreement reflects
   i. different weightings over regions of the joint distribution
   ii. different sensitivity to changing shape (any measure = rank-dependence + distributions)
3. Concordance depends on who gains skill, not how much (Becker-Tomes-Loury)
   - Raising every child's skills can leave mobility untouched

Contributions

1. Unifying framework w/ Concordance: sufficient & necessary for agreement
    - Adds to measure comparisons (Deutscher, Mazumder; Berman), axiomatic approaches (Shorrocks; Fields, Ok; D'Agostino, Dardanoni; Cowell, Flachaire), dual of 2D-inequality problem (Atkinson, Bourguignon; Dardanoni)
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   - Diagnostic tool: agreement among measures $\neq$ robustness
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3. Microfounded w/ incidence in Becker-Tomes-Loury models
   - Heterogeneous effects by family background (Jackson et al.; Chetty et al.; Biasi)
   - Sorting: NAM vs PAM in matching (Anderson, Smith; Gola; Boerma et al.) & mating (Chiappori et al.)

Setup

Copulas and Concordance: Visual Intuition ▸ Example II

Sklar (1959): $F(Y^K, Y^P) = C\big(F_K(Y^K), F_P(Y^P)\big)$ Incomes (Y^K, Y^P), C = joint dist. of ranks

- Copula encodes *all* rank-dependence $\perp\!\!\!\perp$ of income distributions ▸ Mass points

  $\longrightarrow$ positional measures are different summaries of the same copula

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Par. Quintile \ Child's Quintile	1	2	3	4	5
1	0.20	0.20	0.20	0.20	0.20
2	0.20	0.20	0.20	0.20	0.20
3	0.20	0.20	0.20	0.20	0.20
4	0.20	0.20	0.20	0.20	0.20
5	0.20	0.20	0.20	0.20	0.20

$\longrightarrow$

Par. Quintile \ Child's Quintile	1	2	3	4	5
1	0.44	0.26	0.15	0.11	0.04
2	0.20	0.40	0.18	0.13	0.09
3	0.12	0.15	0.45	0.16	0.12
4	0.04	0.11	0.20	0.40	0.25
5	0.20	0.08	0.02	0.20	0.50

$\longrightarrow$

Par. Quintile \ Child's Quintile	1	2	3	4	5
1	1.00	0.00	0.00	0.00	0.00
2	0.00	1.00	0.00	0.00	0.00
3	0.00	0.00	1.00	0.00	0.00
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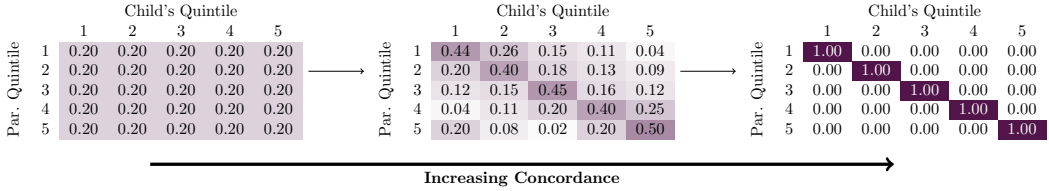
Increasing Concordance

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- Shift mass from off-diagonal (origin independent) to diagonal (rank-preserving) cells

- Elementary exchange $\approx$ Pigou-Dalton transfer for mobility ▸

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

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- Higher concordance $\longleftrightarrow$ **lower mobility**

When Do Measures Agree?

Concordance Orders All Standard Mobility Measures

▸ No time? Skip intuition

Props 1-4: Fix marginals. If $C^A \succeq C^B$, then economy A is *less mobile* than B by:

1. Regression/Correlation-based measures:
   - IGE: $\beta^A \geq \beta^B$, Rank-rank slope: $\rho^A \geq \rho^B$, Intergenerational correlation
2. "Local" measures:
   - Conditional expected rank $\longrightarrow$ rotates around the median (linear fit)
3. Transition probabilities:
   - diagonal $\uparrow$, off-diagonal $\downarrow$ (at every threshold), Directional mobility†
4. Axiomatic measures:
   - Shorrocks trace†, Fields & Ok, D'Agostino & Dardanoni, Cowell & Flachaire

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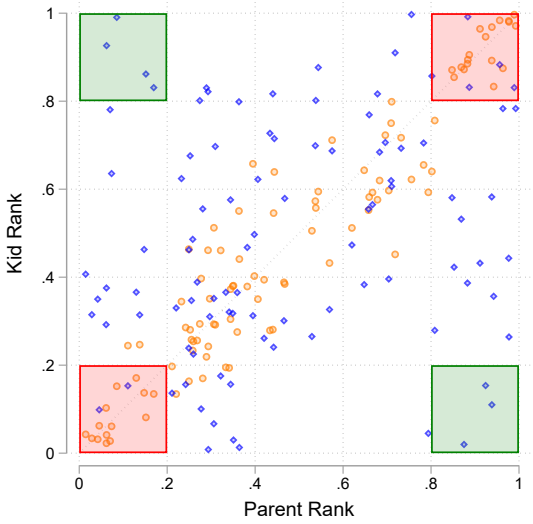
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$\longrightarrow$ Measures are ordinal substitutes: observe one ranking, know all others

† requires stronger diagonal concentration A1 common proxies & measurement error (Prop. 5)

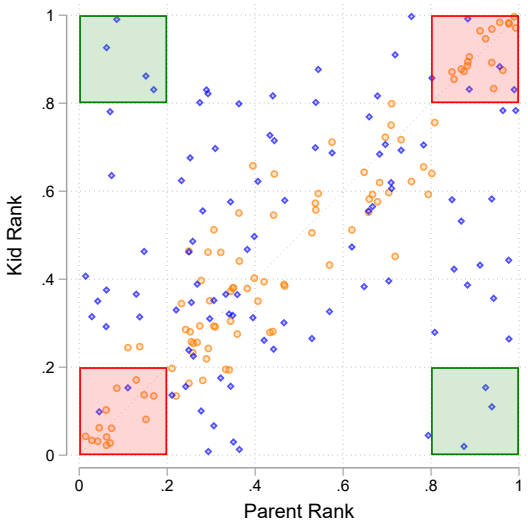
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Transition probability: *one point* of the copula



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Mobility Measure:

$$\Pr(R^K > \tau \mid R^P > \tau) = \frac{1-2\tau+C(\tau,\tau)}{1-\tau}$$ (Q5→Q5: $\tau = 0.8$)

If $C^A \geq C^B$ everywhere

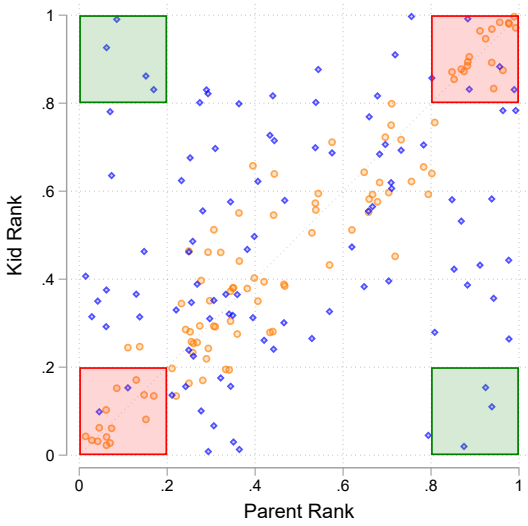
$\longrightarrow C^A(\tau,\tau) \geq C^B(\tau,\tau)$

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(Copula definition, simplest cases!)

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Uses C at *one point*

$\longrightarrow$ depends on where you look

When Concordance Fails, Measures Must Disagree

Concordance is a partial order $\longrightarrow$ not all economies are ranked, e.g.

Par. Quintile \ Child's Quintile	1	2	3	4	5
1	0.32	0.24	0.2	0.15	0.08
2	0.25	0.23	0.21	0.18	0.12
3	0.19	0.22	0.22	0.21	0.16
4	0.15	0.18	0.21	0.24	0.22
5	0.09	0.12	0.16	0.22	0.41

No Concordance Order $\longleftrightarrow$

Par. Quintile \ Child's Quintile	1	2	3	4	5
1	0.58	0.00	0.00	0.14	0.28
2	0.00	0.86	0.00	0.00	0.14
3	0.00	0.00	1.00	0.00	0.00
4	0.14	0.00	0.00	0.86	0.00
5	0.28	0.14	0.00	0.00	0.58

- Identical rank correlation (0.32), but different transition probabilities

When Concordance Fails, Measures Must Disagree

- For fixed marginals, any measure:

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- $\longrightarrow$ Agreement among a few measures $\neq$ robustness

Decomposition: Positional Dependence vs. Marginals ▸ e.g.

- Comparisons across time/space typically involve different marginal distributions

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- Rank-based measures: Marginal component $= 0$ by construction!
- Level-based measures (e.g. IGE): Both components non-zero & potentially opposite

  Order matters $\longrightarrow$ average over both orderings (Shapley-Owen-Shorrocks)

What Moves Concordance?

Microfoundation: Becker–Tomes–Loury

- Classic dynastic investment model

$$\max_{I} \; U\Big(\overbrace{Y^P + T(Y^P) - I}^{\text{parent consumption}}\Big) + \delta\, \mathbb{E}_\theta\Big[U\Big(\overbrace{W\big(\underbrace{\theta f(I, Y^P)}_{H^K}\big)}^{Y^K}\Big)\Big]$$

  - Parents choose investment I; T: transfers; f: skill technology
  - W: wage schedule, strictly increasing θ: luck, realised after investment

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- **Skill index** $S(y) = \ln f\big(I^\star(y), y\big) \longrightarrow \ln H^K = S(Y^P) + \ln\theta$
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- When do changes in technology, preferences, or policy lead to more mobility?

Who Gains Skill, Not How Much

Prop 6: Let incidence $\Delta(y) = S^A(y) - S^B(y)$

$$\Delta \text{ increasing in } y \longrightarrow C^A \succeq C^B$$ (average of Δ irrelevant)

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- Investment ↑ for **every family**: mobility can fall, stay *unchanged*, or *rise* (Cor. 3)
- Return to skill: copula unchanged, IGE scales $\longrightarrow$ ranks track who gains

Who Gains Skill, Not How Much

Prop 6: Let incidence $\Delta(y) = S^A(y) - S^B(y)$

$$\Delta \text{ increasing in } y \longrightarrow C^A \succeq C^B \qquad \text{(average of } \Delta \text{ irrelevant)}$$

- Investment ↑ for **every family**: mobility can fall, stay *unchanged*, or *rise* (Cor. 3)
- Return to skill: copula unchanged, IGE scales $\longrightarrow$ ranks track who gains
- **Cor. 4:** Both tails gain more than average $\longrightarrow$ copulas cross
  - e.g., targeted programs + elite sorting (as across CZs)

U.S. Mobility in the 20th Century

Has Mobility Increased? Testing the Concordance Order Details

- Replicate Jácome, Kuziemko & Naidu (2025)

  15 pooled surveys; parental income imputed from occupation $\times$ race $\times$ region; U.S.-born men 30–50; 56,102 father-son pairs

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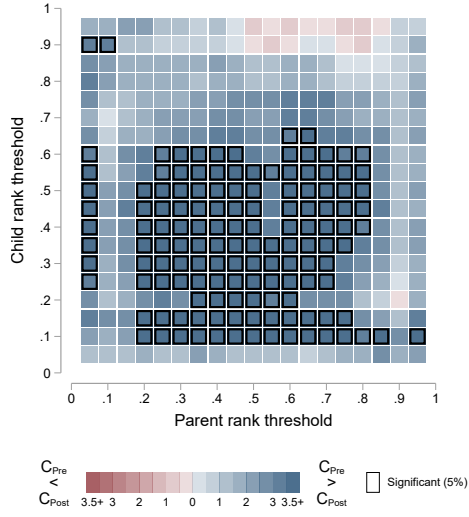
- Successive decades?

Decade A	Decade B	$pval_{A \succeq B}$	$pval_{B \succeq A}$	Interpretation
1920	1930	.469	.178	Indistinguishable
1930	1940	.813	.001	↑ mobility
1940	1950	.001	.262	↓ mobility
1950	1960	.001	.001	Copulas cross
1960	1970	.581	.026	↑ mobility

Long Run: Broad-Based Positional Mobility Gains

1910s vs. 1950s (reject one)

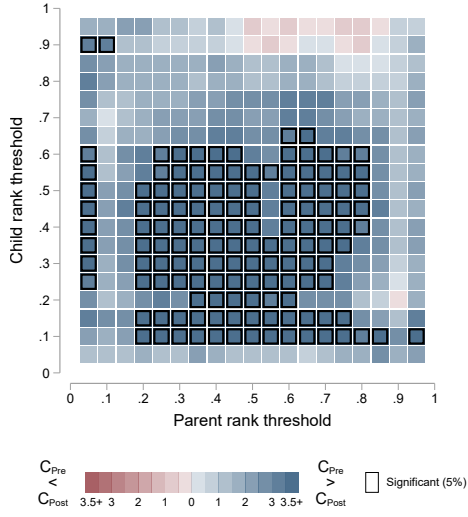
- Each cell: $C^A \lessgtr C^B$
- Blue: later cohort more mobile



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- Broad-based gains



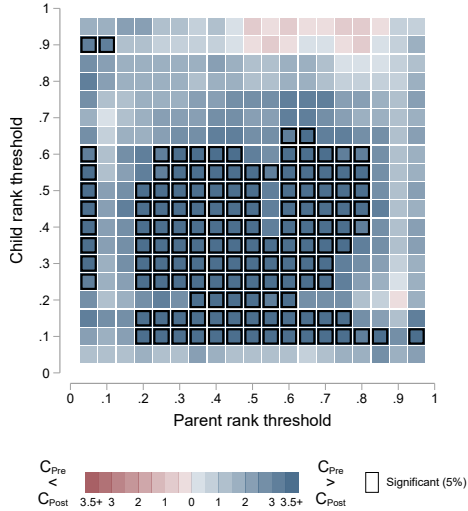
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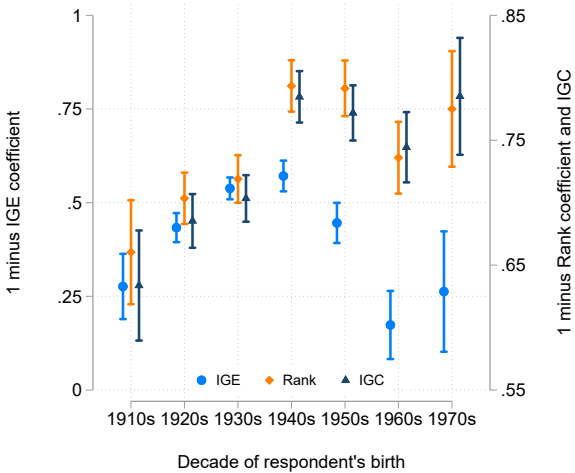
Has U.S. mobility increased?

- Yes! vs Early 20thC
- *Every* rank-based measure
- But not monotone



Long Run: Rank Gains vs. the IGE

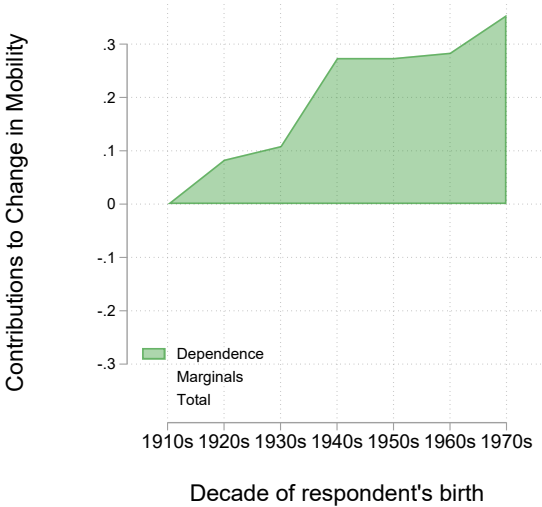
Other Measures



- Rank-rank slope: mobility rises to mid-century, reverses for 1960s
- IGC tracks rank-rank (not generic)
- IGE diverges starting 1950s

Long Run: Rank Gains vs. the IGE

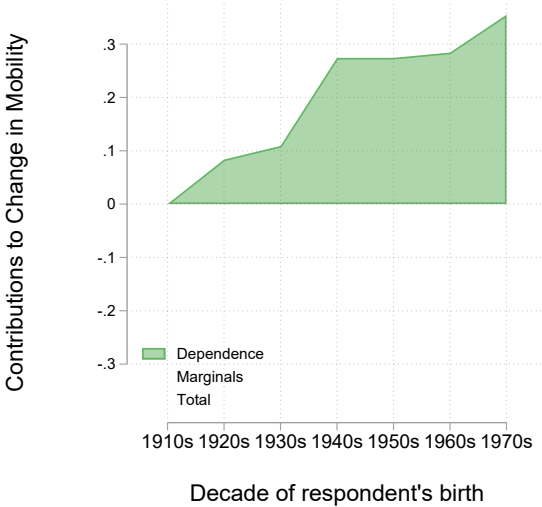
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- **Dependence component:** rising mobility + plateau
  - Like the rank-rank slope

Long Run: Rank Gains vs. the IGE

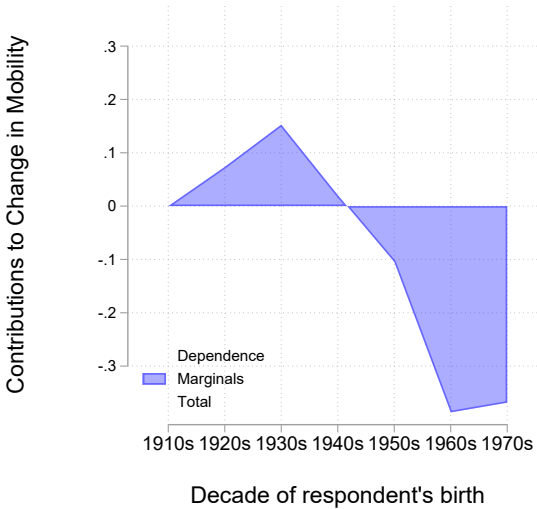
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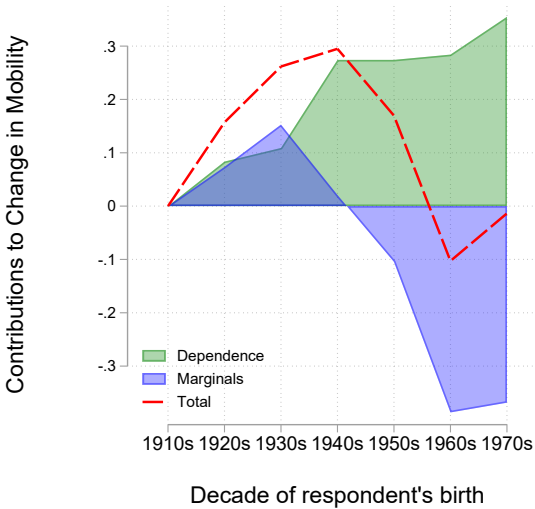
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- **Dependence component:** rising mobility + plateau
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- **Marginal component:** child vs. parent dispersion pulls mobility down
  - $\frac{\sigma_K^{1960}/\sigma_P^{1960}}{\sigma_K^{1910}/\sigma_P^{1910}} \approx 1.75$

Long Run: Rank Gains vs. the IGE

Other Measures



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  - $\frac{\sigma_K^{1960}/\sigma_P^{1960}}{\sigma_K^{1910}/\sigma_P^{1910}} \approx 1.75$
- Dominates by 1960s $\longrightarrow$ post-1950 IGE mostly marginals

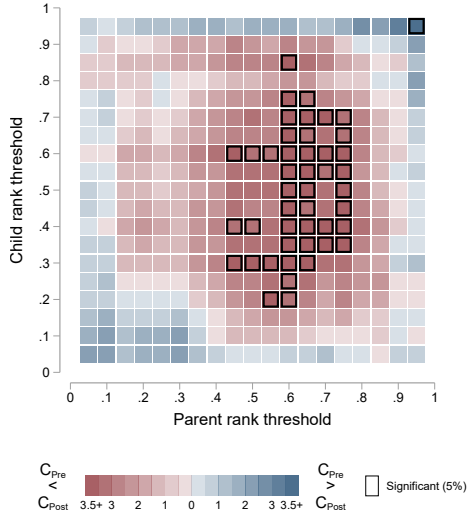
Mid-Century: Where the Copulas Cross

Other Measures

1950s vs. 1960s (reject both)

- Mostly red: falling mobility

Skip



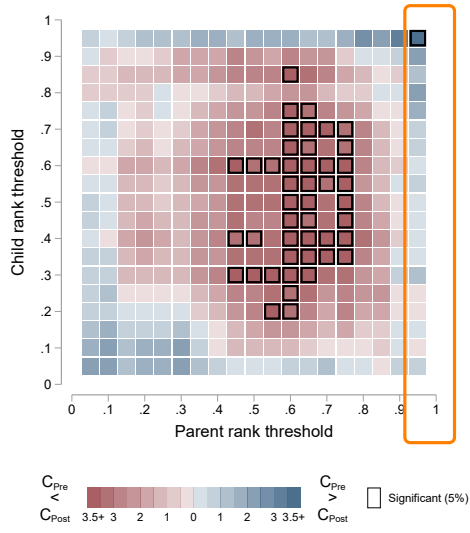
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- Top ventile: mobility rises

Skip



Mid-Century: Where the Copulas Cross

▸ Other Measures

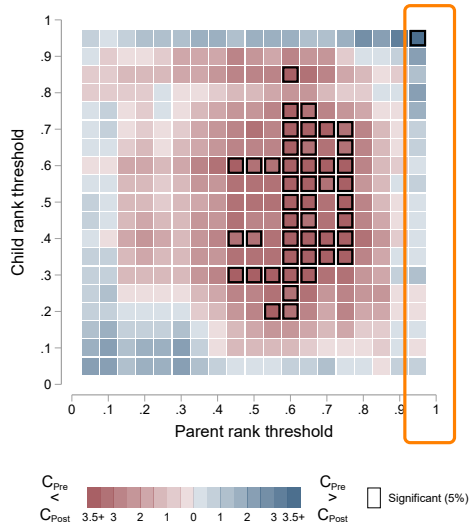
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Declining mobility for the 1960s cohort?

- Measure specific (region weights)

▸ Skip



Mid-Century: Where the Copulas Cross

▸ Other Measures

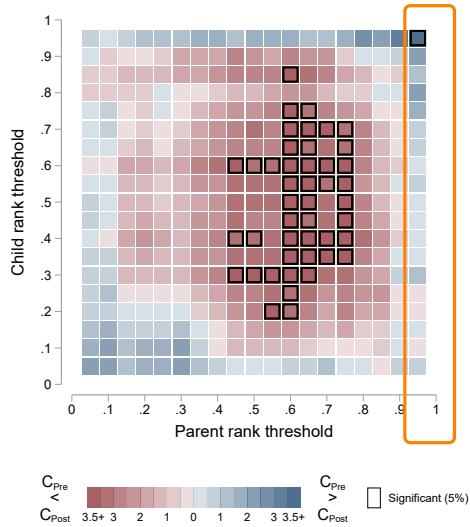
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▸ Skip



Mid-Century: Where the Copulas Cross

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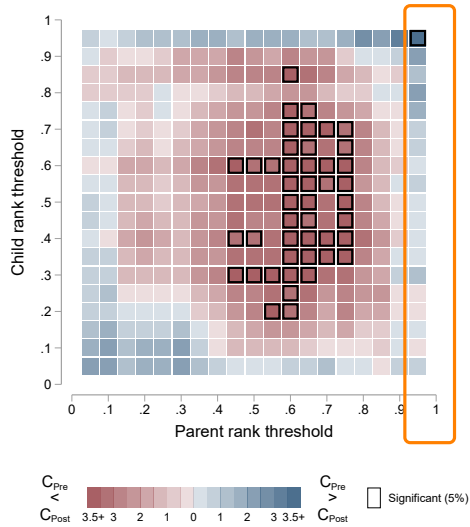
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▸ Skip



Mid-Century: Where the Copulas Cross

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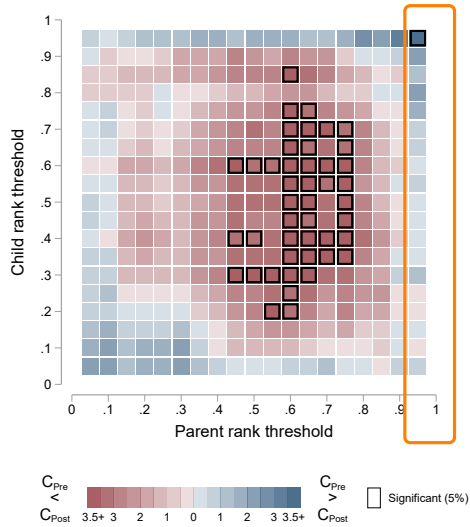
Declining mobility for the 1960s cohort?

- Measure specific (region weights)
- Period of tension in existing work
- But appears “robust”

Model ⟶ non-monotone incidence

rise with resources, stall at the top
e.g., GI Bill? (Stanley, 2003)

▸ Skip



Conclusions

Implications For Applied Researchers

A single property—concordance—determines when standard mobility measures agree

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A single property—concordance—determines when standard mobility measures agree

For applied researchers:

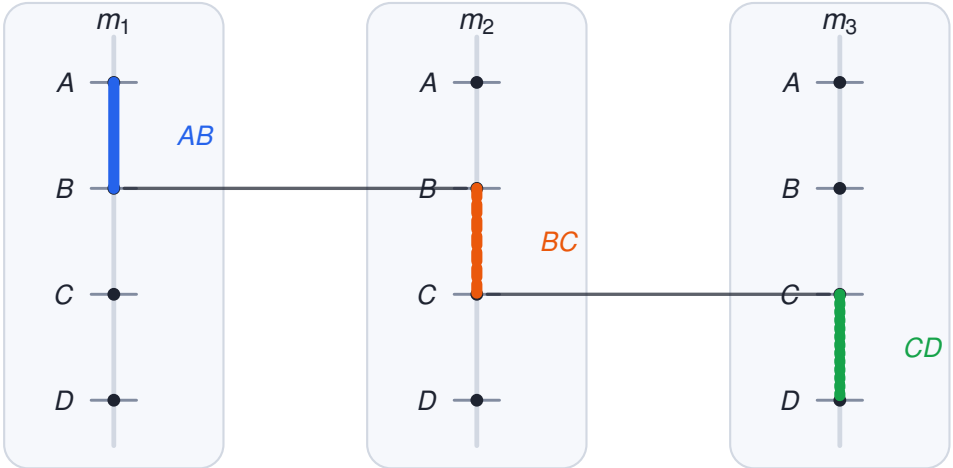
1. When concordance holds: measures are substitutes
   - Pick by data quality, not axiom: transitions for noisy data, IGE for clean panels, rank-rank for cross-country

2. When it fails: disagreement is informative, not noise
   - Agreement among a few measures $\neq$ robustness
   - Test it; locate it; report local rank-rank slopes & transition probabilities

3. Decomposition as diagnostic: rank-dependence vs. changing marginals

4. Policy: who gains, not how much $\longrightarrow$ effects across the full parental distribution

Chaining Measures



Gives full comparison of A, B, C and D in *any* measure

Details

Mass Points

Back

e.g. from bunching, top-coding, censoring, genuine discreteness $\longrightarrow$ no restrictions on F_K or F_P, but Copula 'indeterminate'

Solutions:

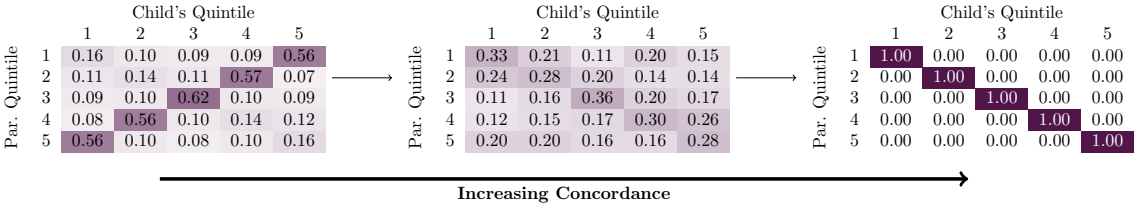
1. Parametric copulas
   - Does violence to discrete data
   - Restrictive

2. Carley (2002) bounds on copulas
   - A lot more math, but flexible
   - Worst-case bounds may not be tight

3. Pick an extension: checkerboard or Piecewise bilinear
   - Propositions 11, 13 in Genest and Nešlehová (2007) show the unique extension preserving concordance order of the original joint-distributions

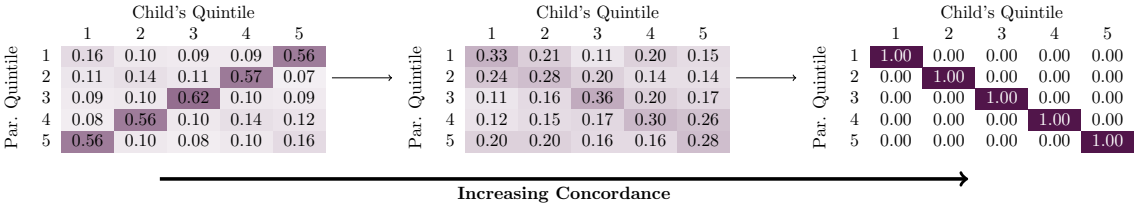
Concordance: Visual Intuition

Back



Concordance: Visual Intuition

Back



- Not just extreme cases

Assumption 1: Strong Diagonal Concentration

Back

Two results need more than concordance: mass must move toward the diagonal

1. Directional contraction: $\exists\, r^\star$ such that, for a.e. r and $s \geq 0$,

$$C^A_{K|P}(r+s \mid r) \geq C^B_{K|P}(r+s \mid r) \quad (r \leq r^\star), \qquad C^A_{K|P}(r-s \mid r) \leq C^B_{K|P}(r-s \mid r) \quad (r \geq r^\star)$$

   - Fewer large moves up from the bottom, down from the top $\longrightarrow$ Directional mobility

2. Diagonal interval dominance: for every rank interval $(a, b]$,

$$\Pr^A\Big(a < R^K \leq b,\ a < R^P \leq b\Big) \geq \Pr^B\Big(a < R^K \leq b,\ a < R^P \leq b\Big)$$

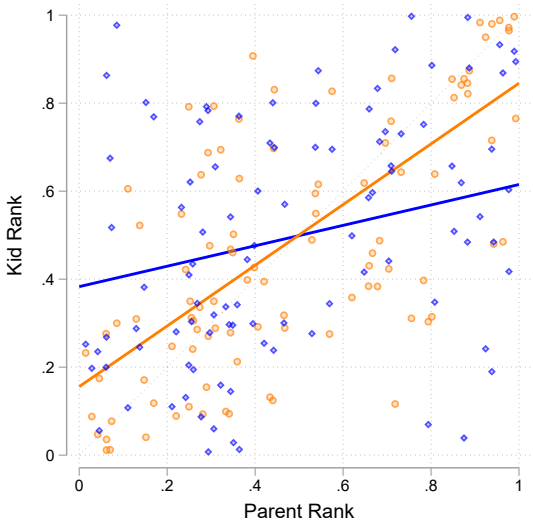
   - More parent-child pairs stay in the same interval $\longrightarrow$ Shorrocks trace

Both hold for Gaussian and FGM copulas as dependence rises. Pointwise (non-parametric) CER rotation instead needs $G(r) = \int_0^1 [C^A(u,r) - C^B(u,r)]\, du$ single-peaked at $r^\star$; the linear fit rotates around the median with no extra assumptions.

Why Does Concordance Order These Measures?

Back

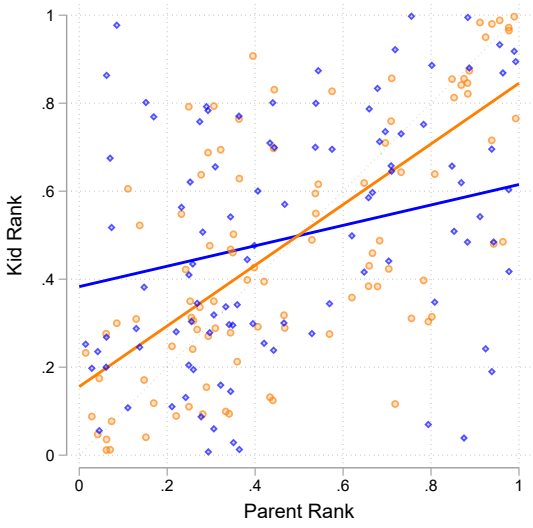
Rank-rank slope: an *average* of the copula



Why Does Concordance Order These Measures?

Back

Rank-rank slope: an *average* of the copula



Mobility Measure:

$$\rho = 12 \int\int \left[C(u,v) - uv \right] du\, dv$$

$C^A \geq C^B$ everywhere

$$\longrightarrow \int\int C^A \geq \int\int C^B$$

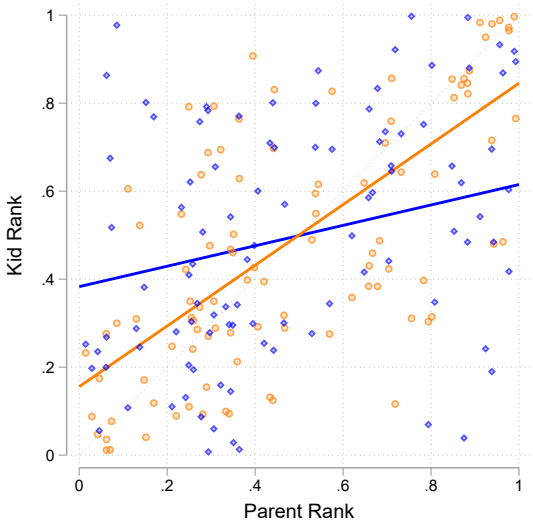
$$\longrightarrow \rho^A \geq \rho^B$$

(Lehmann 1966; Tchen 1980)

Why Does Concordance Order These Measures?

Back

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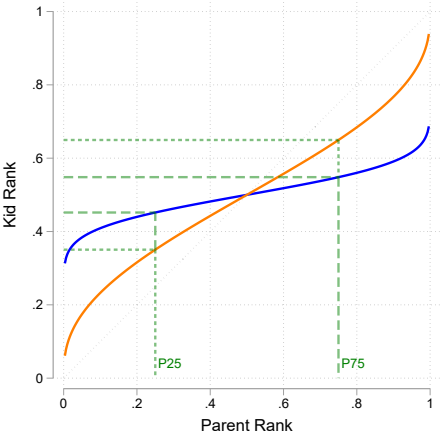
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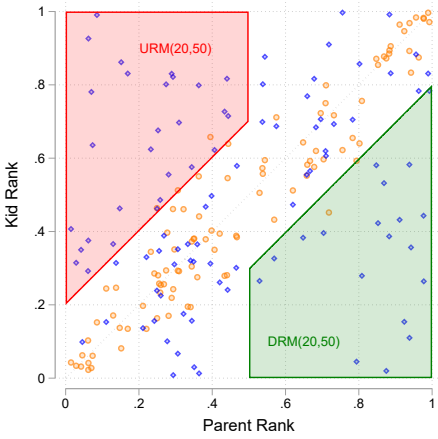
(Lehmann 1966; Tchen 1980)

Uses *C everywhere*
$\longrightarrow$ regions can offset

Why Does Concordance Order These Measures? Illustration III

Back





Extension: Proxies & Measurement Error

Back

Prop 5: Observe proxy $E^g = H_g(R^g, \varepsilon^g)$ for generation g, H_g increasing, and ε independent of ranks. Let $\widetilde{C}$ denote copula for (E^K, E^P). Concordance ordering preserved:

$$C^A \succeq C^B \quad \longrightarrow \quad \widetilde{C}^A \succeq \widetilde{C}^B$$

Extension: Proxies & Measurement Error

Back

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$\longrightarrow$ Error may bias *levels* but ordering *preserved*

Extension: Proxies & Measurement Error

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- Covers common proxy cases

1. Education/occupation as income proxies: $Y \longrightarrow E$
2. Income and Education as skill proxies: $S \longrightarrow E, Y$

Extension: Proxies & Measurement Error ◂ Back

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- Covers common proxy cases ▸
  1. Education/occupation as income proxies: $Y \longrightarrow E$
  2. Income and Education as skill proxies: $S \longrightarrow E, Y$
- Covers flexible measurement error cases $\widetilde{Y} = E = H_g(R^g, \varepsilon^g)$ (Corollary 1) ▸
  1. (non-)Classical measurement error in (log-)incomes (Bound, Brown, Duncan & Rogers 1994)
  2. Measurement error in ranks (Chetverikov & Wilhelm 2023)
  3. Life cycle biases (Kitagawa, Nybom, & Stuhler 2018)

Proxies & Meas Error: Functional Forms

Back

Proxies:

1. Human capital determines incomes $Y = H(E) + \varepsilon$ with $E = H^{-1}(Y - \varepsilon)$
2. Ability determines education and income $E = H_E(S, \varepsilon)$ and $Y = H_Y(S, \varepsilon)$.

Proxies & Meas Error: Functional Forms

Back

Proxies:

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Measurement error:

1. Classical measurement error in incomes
   $\widetilde{Y}^K = F_K^{-1}(R^K) + \varepsilon^K, \qquad \widetilde{Y}^P = F_P^{-1}(R^P) + \varepsilon^P$

2. Measurement error in ranks
   $\widetilde{R}^K = F_{\widetilde{K}}\left(F_K^{-1}(R^K) + \varepsilon^K\right), \qquad \widetilde{R}^P = F_{\widetilde{P}}\left(F_P^{-1}(R^P) + \varepsilon^P\right)$

3. Life Cycle Biases
   $\widetilde{Y}^K = G_K\left(F_K^{-1}(R^K) + \mu_{\varepsilon^K} + \sigma_{\varepsilon^K}\varepsilon^K\right), \qquad \widetilde{Y}^P = G_P\left(F_P^{-1}(R^P) + \mu_{\varepsilon^P} + \sigma_{\varepsilon^P}\varepsilon^P\right)$

Measure specific weights

Back

- Measures expressed as

$$M_M = \int \int \phi_M \left(Y^K, Y^P \right) f \left(Y^K, Y^P \right) d\, Y^K d\, Y^P,$$

where $\phi_M \left(Y^K, Y^P \right)$: mobility-measure specific intensity

- For fixed marginals, any twice-differentiable measure: $M_M^A - M_M^B$

$$= \int_0^1 \int_0^1 W_M(R^K, R^P; F^K, F^P) \left[C^A(R^K, R^P) - C^B(R^K, R^P) \right] dR^K\, dR^P$$

where

$$W_M(R^K, R^P) = \frac{\phi_{M,12}\left(F_K^{-1}(R^K), F_P^{-1}(R^P)\right)}{f_K \left(F_K^{-1}(R^K) \right)\ f_P \left(F_P^{-1}(R^P) \right)}$$

- Works for non-continuous cases also

Jácome, Kuziemko & Naidu (2025) Replication

Back

15 Pooled Surveys:

- General Social Survey (GSS), American National Election Survey (ANES)
- Panel Study of Income Dynamics (PSID), Occupational Changes in a Generation (OCG)
- National Longitudinal Surveys (NLS): Mature Women, Older Men, Younger Women
- National Survey of Black Americans, National Fertility Survey, others

Sample Construction:

- Inclusion: US-born sons, ages 30–50, non-missing family income + father's occupation
- Race: White and Black only (other races $< 1\%$ pre-1950)
- Weights: Reweighted to census race shares per birth decade

Parental Income Imputation:

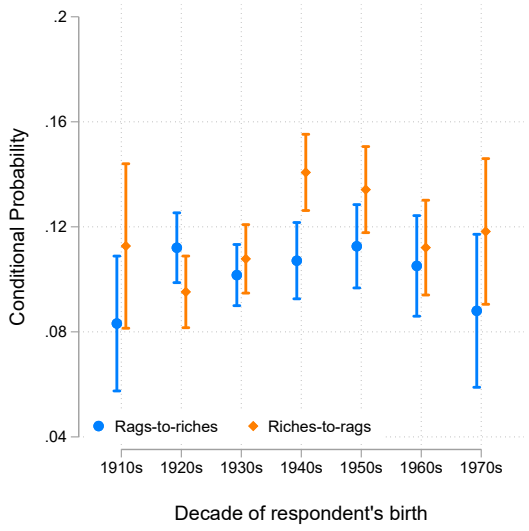
- Predict $\widehat{Y}^P$ using 28 harmonized occupation categories $\times$ race $\times$ South
- Sources: 1901 Cost of Living Survey, 1936 Expenditure Survey, 1940–90 Censuses
- Matched to respondent's $\approx$10th birthday; special adjustments for farmers

Sample Sizes by Birth Decade:

1910s	1920s	1930s	1940s	1950s	1960s	1970s
4,641	11,857	11,426	10,288	9,531	5,691	2,668

Transition Matrix Measures Co-Move

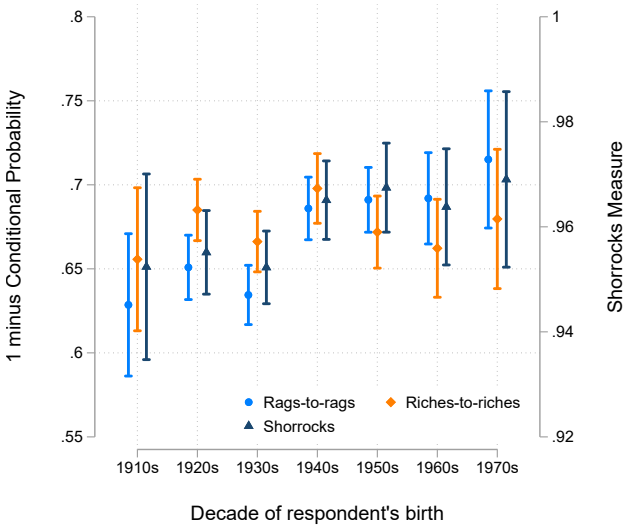
Back



- Quintile Transition Matrix
- Q1→Q5 & Q5→Q1

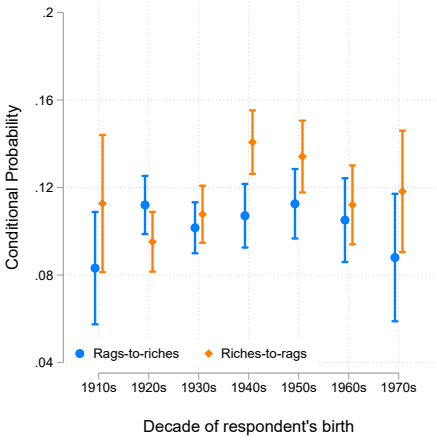
Transition Matrix Measures Co-Move

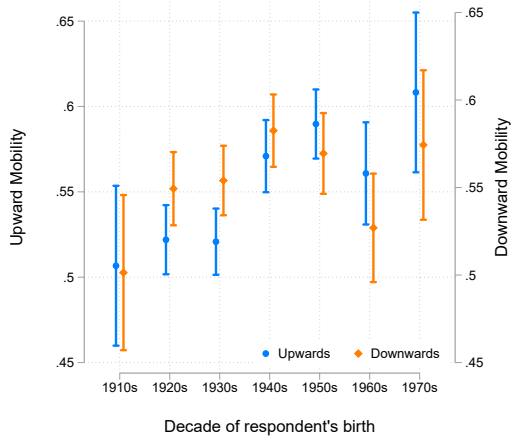
Back



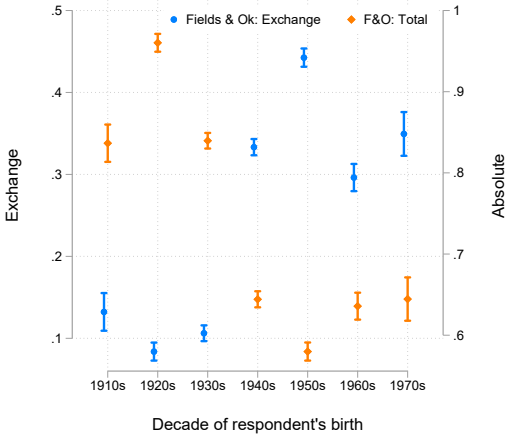
- Quintile Transition Matrix
- Q5→Q5 & Q1→Q1 & Trace

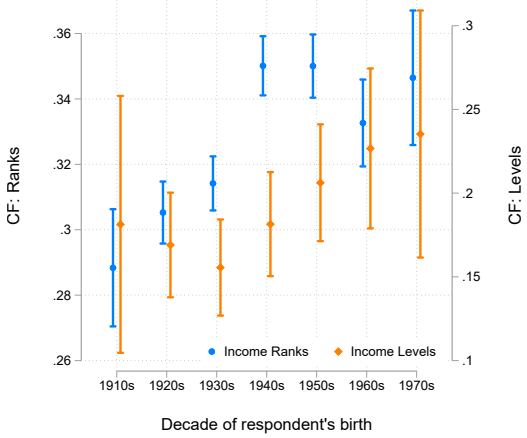
Back


.2
.16
.12
.08
.04
Conditional Probability
Rags-to-riches
Riches-to-rags
1910s
1920s
1930s
1940s
1950s
1960s
1970s
Decade of respondent's birth

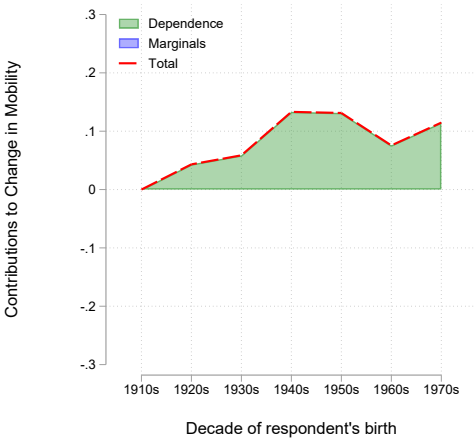

.65
.6
.55
.5
.45
Upward Mobility
Upwards
Downwards
Downward Mobility
1910s
1920s
1930s
1940s
1950s
1960s
1970s
Decade of respondent's birth

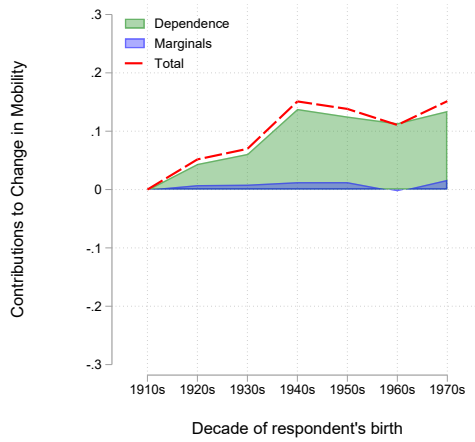
Back


Fields & Ok: Exchange
F&O: Total
Exchange
Absolute
.5
.4
.3
.2
.1
1
.9
.8
.7
.6
1910s
1920s
1930s
1940s
1950s
1960s
1970s
Decade of respondent's birth

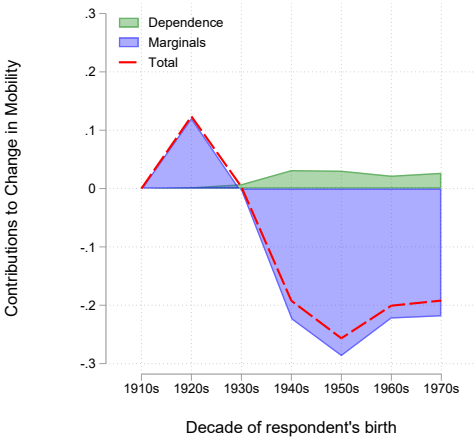

CF: Ranks
CF: Levels
.36
.34
.32
.3
.28
.26
.3
.25
.2
.15
.1
Income Ranks
Income Levels
1910s
1920s
1930s
1940s
1950s
1960s
1970s
Decade of respondent's birth

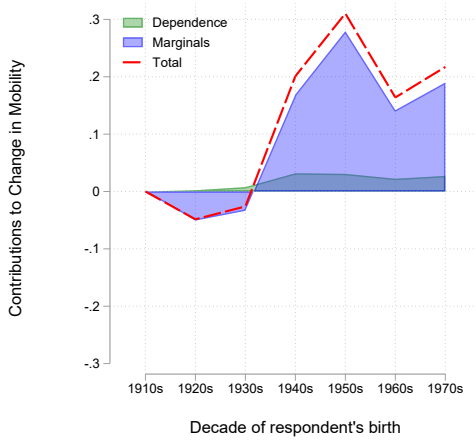
Back


Dependence
Marginals
Total
.3
.2
.1
0
-.1
-.2
-.3
Contributions to Change in Mobility
1910s
1920s
1930s
1940s
1950s
1960s
1970s
Decade of respondent's birth


Dependence
Marginals
Total
.3
.2
.1
0
-.1
-.2
-.3
Contributions to Change in Mobility
1910s
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Back


Dependence
Marginals
Total
.3
.2
.1
0
-.1
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-.3
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Marginals
Total
.3
.2
.1
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-.1
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-.3
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1910s
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Testing The Concordance Order

Back

Set up as a moment inequality testing procedure (Andrews and Soares, 2010)

- Moments on a grid $\mathcal{G}$:

$$m_g \;=\; C_A(u_g, v_g) - C_B(u_g, v_g), \qquad g = 1, \ldots, G.$$

- Null hypothesis of dominance

$$H_0 : \; m_g \geq 0 \;\text{ for all } g = 1, \ldots, G \qquad \text{vs.} \qquad H_1 : \; m_g < 0 \;\text{ for some } g.$$

- Test statistic is the minimum of the studentized moments:

$$T_n \;=\; \min_{g=1,\ldots,G} \frac{\widehat{m}_g}{\widehat{s}_g},$$

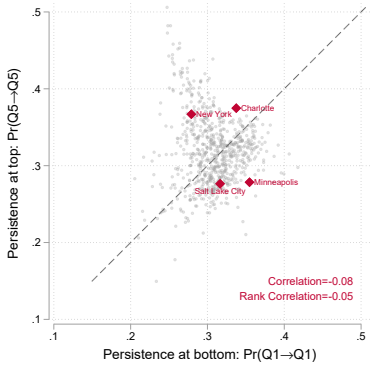
  w/ bootstrap standard error $\widehat{s}_g$

- Test $A \succeq B$ and $B \succeq A$

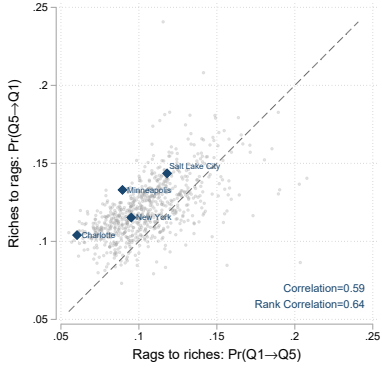
Spatial Mobility: Measure Disagreement Across CZs

Back

Chetty et al. (2014) CZ transition matrices, remapped to within-CZ copulas



Q1→Q1 vs. Q5→Q5 persistence



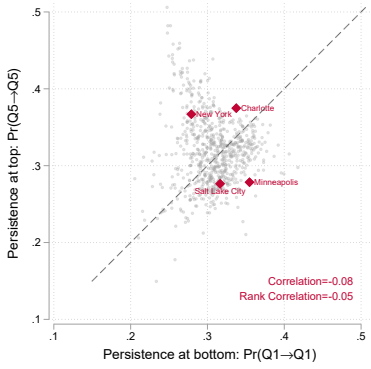
Q1→Q5 vs. Q5→Q1 tail mobility

- Salt Lake $\succ$ Charlotte on *every* measure...
  - ...but Salt Lake vs. NYC, NYC vs. Minneapolis: depends on measure

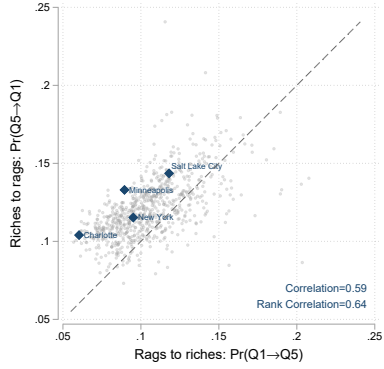
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Q1→Q5 vs. Q5→Q1 tail mobility

- Salt Lake $\succ$ Charlotte on *every* measure...
  - ...but Salt Lake vs. NYC, NYC vs. Minneapolis: depends on measure
- Persistence top-quartile$\neq$ in **91%** of CZs; **61%** for the tail-mobility

A Simple Illustrative Example

Back

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N\left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \quad \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix}\right]$$

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What do measures depend on?

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- Rank-rank slope: $\text{Corr}(R^K, R^P) = {}^6/_\pi \arcsin(\rho/2) \approx \rho$ (Kruskal, 1958) *only* dependence!

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- IGE: $\beta = \rho \cdot \frac{\sigma_K}{\sigma_P}$ dependence *and* inequality
  - Changes: $\Delta\beta = \underbrace{(\rho^B - \rho^A) \cdot \frac{\sigma_K^B}{\sigma_P^B}}_{\text{change in rank correlation}} + \underbrace{\rho^A \cdot \left(\frac{\sigma_K^B}{\sigma_P^B} - \frac{\sigma_K^A}{\sigma_P^A} \right)}_{\text{change in inequality}}$

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  - Rising inequality ($\sigma_K/\sigma_P \uparrow$) can dominate IGE even as rank dependence weakens ($\rho \downarrow$)

Proof Sketch: Incidence $\longrightarrow$ Concordance

Back

- Path between economies: $S_z = S^B + z\Delta,\ z \in [0,1]$, with copula C_z $(C_0 = C^B, C_1 = C^A)$

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- Fix child rank u and parental rank v; split parents into L (below v) and H (above):

$$\frac{\partial C_z(u,v)}{\partial z} = \underbrace{\frac{W_L W_H}{W_L + W_H}}_{>0} (\bar{\Delta}_H - \bar{\Delta}_L)$$

$\bar{\Delta}_J$: average incidence among group-J parents of children at rank u (weighted by the density of luck)

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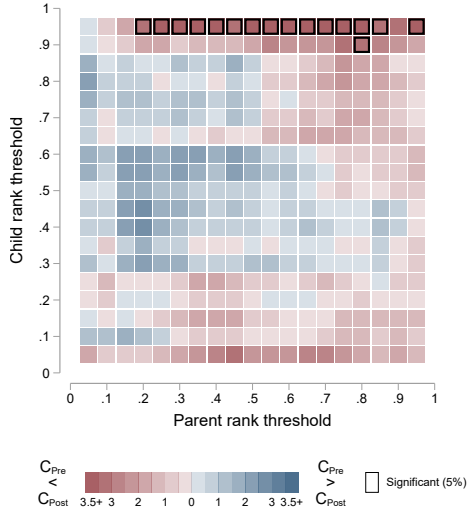
- Δ nondecreasing $\longrightarrow \bar{\Delta}_H \geq \bar{\Delta}_L$ everywhere $\longrightarrow C_z$ rises at every $(u,v) \longrightarrow C^A \succeq C^B$
- Constant shift in Δ: every child's log skill moves equally $\longrightarrow$ ranks, and copula, unchanged
- W strictly increasing $\longrightarrow$ income ranks $=$ skill ranks for *any* W

Mid-Century: Rising Top Persistence

Back

1940s vs. 1950s (reject one)

- 1950s cohort less mobile
- Driven by ↑ persistence high-ranked children
- Insignificant elsewhere



Mid-Century: Rising Top Persistence

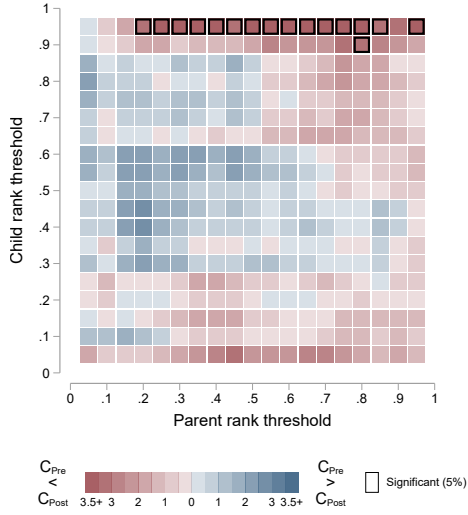
Back

1940s vs. 1950s (reject one)

- 1950s cohort less mobile
- Driven by ↑ persistence high-ranked children
- Insignificant elsewhere

⟶ top persistence peaks for 50s cohort

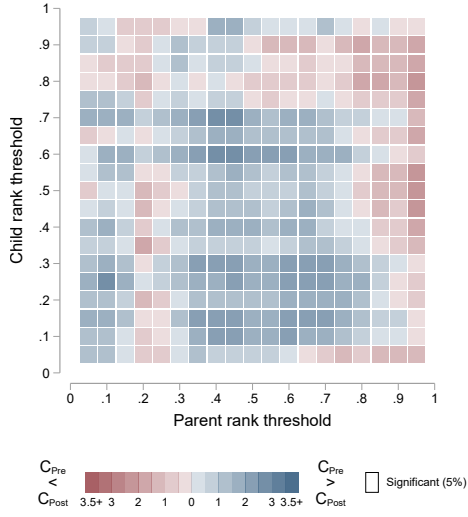
⟶ Rank-Rank essentially flat



1920s vs. 1930s: Indistinguishable

Back

- Neither direction rejected
- Pattern resembles a crossing, but violations are small



Concordance is Pigou-Dalton for Mobility

Back

	Inequality (Atkinson, 1970)
Elementary operation	Pigou-Dalton transfer
Partial order	Lorenz dominance (SOSD)
Sufficiency	Inequality measures agree

Analogues in SWF for 2D ineq. (Atkinson-Bourguignon) & Perm. Income ineq. (Dardanoni)

Concordance is Pigou-Dalton for Mobility

Back

	Inequality (Atkinson, 1970)	*Mobility* (Today)
Elementary operation	Pigou-Dalton transfer	Concordance exchange
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Sufficiency	Inequality measures agree	Mobility measures agree

Exchange: swap two children's incomes so the richer child has the richer parent $\longrightarrow$ concordance $\uparrow$

Analogues in SWF for 2D ineq. (Atkinson-Bourguignon) & Perm. Income ineq. (Dardanoni)

Concordance is Pigou-Dalton for Mobility

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	Inequality (Atkinson, 1970)	*Mobility* (Today)
Elementary operation	Pigou-Dalton transfer	Concordance exchange
Partial order	Lorenz dominance (SOSD)	Concordance
Sufficiency	Inequality measures agree	Mobility measures agree
When it *fails*	Lorenz curves cross	Copulas cross

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